

SKILL WEEK – 3

Session – 5

TASK 1: TO Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases



```
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 200
#define MAX_QUOTES 20

int main() {
    char input[MAX_INPUT];
    char quoted_strings[MAX_QUOTES][MAX_INPUT];
    int quote_count = 0;

    printf("Session 5 - Single Quote Parsing\n");
    printf("Enter a command with single quotes:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    int i = 0;

    while (input[i] != '\0') {
        if (input[i] == '\'' ) {
            i++;

            int j = 0;

            while (input[i] != '\0' && input[i] != '\0') {
                if (j < MAX_INPUT - 1) {
                    quoted_strings[quote_count][j++] = input[i];
                }
                i++;
            }

            quoted_strings[quote_count][j] = '\0';

            if (input[i] == '\0') {
                quote_count++;
                i++;
            } else {
                printf("Error: Unclosed single quote.\n");
                return 1;
            }
        }
    }
}
```

```

        if (j < MAX_INPUT - 1) {
            quoted_strings[quote_count][j++] = input[i];
        }
        i++;
    }

    quoted_strings[quote_count][j] = '\0';

    if (input[i] == '\\') {
        quote_count++;
        i++;
    } else {
        printf("Error: Unclosed single quote.\n");
        return 1;
    }
} else {
    i++;
}

printf("\nQuoted strings found:\n");

if (quote_count == 0) {
    printf("No single-quoted strings found.\n");
} else {
    for (int k = 0; k < quote_count; k++) {
        printf("Quoted String %d: [%s]\n",
            k + 1, quoted_strings[k]);
    }
}

printf("\nParsing validation:\n");

if (quote_count > 0)
    printf("Single-quote parsing successful.\n");
else
    printf("No quoted content to validate.\n");

return 0;
}

```

```

keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ nano session5_task1.c
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ gcc session5_task1.c -o session5_task1
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ ./session5_task1

```

```

Session 5 - Single Quote Parsing
Enter a command with single quotes:
myshell> echo 'Hello $USER'

```

```

Quoted strings found:
Quoted String 1: [Hello $USER]

```

```

Parsing validation:
Single-quote parsing successful.

```

```

keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ |

```

TASK 2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands

```
keerthi_priya@LAPTOP-OSGV x + v
GNU nano 8.7.1
#include <stdlib.h>
#include <string.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 300
#define MAX_TOKENS 50
#define MAX_TOKEN_LENGTH 100

void expand_variable(char *input, char *output) {
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < MAX_INPUT - 1) {
        if (input[i] == '$') {
            i++;
            char variable[50];
            int k = 0;

            while (isalnum((unsigned char)input[i]) || input[i] == '_') {
                if (k < 49)
                    variable[k++] = input[i];
                i++;
            }

            variable[k] = '\0';
            char *value = getenv(variable);

            if (value != NULL) {
                int v = 0;

                while (value[v] != '\0' && j < MAX_INPUT - 1) {
                    output[j++] = value[v++];
                }
            } else {
                output[j++] = input[i++];
            }
        } else {
            output[j++] = input[i++];
        }
    }
}
```

```
keerthi_priya@LAPTOP-OSGV x + v
GNU nano 8.7.1
    }
}

output[j] = '\0';
}

int main() {
    char input[MAX_INPUT];
    char expanded[MAX_INPUT];
    char tokens[MAX_TOKENS][MAX_TOKEN_LENGTH];

    int token_count = 0;
    int i = 0;

    printf("Session 5 - Double Quote Parsing\n");
    printf("Double quotes preserve spaces and allow variable expansion.\n");
    printf("Enter a command:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Empty command.\n");
        return 0;
    }

    /* Parse input */
    while (input[i] != '\0' && token_count < MAX_TOKENS) {
        while (isspace((unsigned char)input[i])) {
            i++;
        }

        if (input[i] == '\0')
            break;

        int j = 0;

        /* Double-quoted string */
        if (input[i] == '"') {
            i++;

            while (input[i] != '\0' && input[i] != '"') {
                if (j < MAX_TOKEN_LENGTH - 1)

```

```
keerthi_priya@LAPTOP-OSGV X + v
GNU nano 8.7.1
    if (input[i] == '"') {
        i++;
    } else {
        printf("Error: Unclosed double quote.\n");
        return 1;
    }

    tokens[token_count][j] = '\0';
    token_count++;
}

/* Normal token */
else {
    while (input[i] != '\0' &&
           !isspace((unsigned char)input[i]) &&
           input[i] != '"') {

        if (j < MAX_TOKEN_LENGTH - 1)
            tokens[token_count][j++] = input[i];

        i++;
    }

    tokens[token_count][j] = '\0';

    if (j > 0)
        token_count++;
}

printf("\nTokens before variable expansion:\n");
for (int k = 0; k < token_count; k++) {
    printf("Token %d: [%s]\n", k + 1, tokens[k]);
}

printf("\nVariable expansion:\n");
for (int k = 0; k < token_count; k++) {
    expand_variable(tokens[k], expanded);

    printf("Token %d: [%s]\n", k + 1, expanded);
}

printf("\nParsing validation:\n");
printf("Double-quote parsing successful.\n");
printf("Spaces inside quotes were preserved.\n");
printf("Variable expansion was processed.\n");

return 0;
}
```

```
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ nano session5_task2.c
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ gcc session5_task2.c -o session5_task2
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ ./session5_task2
Session 5 - Double Quote Parsing
Double quotes preserve spaces and allow variable expansion.
Enter a command:
myshell> echo "Hello $USER"

Tokens before variable expansion:
Token 1: [echo]
Token 2: [Hello $USER]

Variable expansion:
Token 1: [echo]
Token 2: [Hello keerthi_priya]

Parsing validation:
Double-quote parsing successful.
Spaces inside quotes were preserved.
Variable expansion was processed.
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ |
```

Session – 6

TASK 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs

```
keerthi_priya@LAPTOP-OSGV × + v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 300

int main() {
    char input[MAX_INPUT];
    | char output[MAX_INPUT];

    int i = 0;
    int j = 0;
    int escaped = 0;

    printf("Session 6 - Escape Sequence Parsing\n");
    printf("Enter a command with escaped spaces or special characters:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';

    if (strlen(input) == 0) {
        printf("Empty input.\n");
        return 0;
    }

    while (input[i] != '\0' && j < MAX_INPUT - 1) {
        if (escaped) {
            output[j++] = input[i];
            escaped = 0;
        } else if (input[i] == '\\') {
            escaped = 1;
        } else {
            output[j++] = input[i];
        }

        i++;
    }

    if (escaped) {
        printf("\nWarning: Escape character at the end of input.\n");
    }

    output[j] = '\0';

    printf("\nOriginal input:\n");
    printf("%s\n", input);

    printf("\nProcessed input:\n");
    printf("%s\n", output);

    printf("\nParser validation:\n");
    printf("Escape sequences processed successfully.\n");
    printf("Escaped characters were preserved.\n");

    return 0;
}
```

```

keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ nano session6_task1.c
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ gcc session6_task1.c -o session6_task1
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ ./session6_task1
Session 6 - Escape Sequence Parsing
Enter a command with escaped spaces or special characters:
myshell> hello\ world

Original input:
hello\ world

Processed input:
hello world

Parser validation:
Escape sequences processed successfully.
Escaped characters were preserved.
keerthi_priya@LAPTOP-OSGVCE7A:/mnt/c/JavaPrograms$ |

```

TASK 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching



```

GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;
    int status;

    printf("Session 6 - Process Execution\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        /* Child process */
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("\nExecuting: ls -l\n");

        execlp("ls", "ls", "-l", NULL);

        /* This runs only if exec fails */
        perror("exec failed");
        exit(EXIT_FAILURE);
    }
    else {
        /* Parent process */
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("\nChild exited with status: %d\n",
                WEXITSTATUS(status));
        }
        else {
            printf("\nChild terminated abnormally.\n");
        }

        printf("Parent process completed.\n");
    }

    return 0;
}

```

```

keerthi_priya@LAPTOP-OSGV  ×  +  ▾
keerthi_priya@LAPTOP-OSGVCE7A: /mnt/c/JavaPrograms$ nano session6_task2.c
keerthi_priya@LAPTOP-OSGVCE7A: /mnt/c/JavaPrograms$ gcc session6_task2.c -o session6_task2
keerthi_priya@LAPTOP-OSGVCE7A: /mnt/c/JavaPrograms$ ./session6_task2
Session 6 - Process Execution

```

```

Parent Process
Child Process
Parent PID: 1498
Child PID: 1499
Child PID: 1499
Parent PID: 1498
Parent waiting for child...

```

```

Executing: ls -l
total 292
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1609 Oct 14 2025 Arithmetic.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 737 Oct 14 2025 Arithmetic.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 727 Oct 14 2025 Arithmetic.txt
-rwxrwxrwx 1 keerthi_priya keerthi_priya 212 Dec 1 2025 Array_output.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1639 Oct 23 2025 Calci.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 954 Oct 23 2025 Calci.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1754 Oct 13 2025 Datatypes.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 639 Oct 13 2025 Datatypes.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 417 Oct 9 2025 Demo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 102 Oct 9 2025 Demo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1062 Oct 16 2025 Divmod.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 326 Oct 16 2025 Divmod.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 529 Oct 15 2025 Edowhile.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 237 Oct 15 2025 Edowhile.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 525 Oct 15 2025 Efor.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 222 Oct 15 2025 Efor.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 231 Oct 15 2025 Efor.txt
-rwxrwxrwx 1 keerthi_priya keerthi_priya 682 Oct 14 2025 Equal.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 354 Oct 14 2025 Equal.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 682 Oct 13 2025 Even.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 527 Oct 15 2025 Ewhile.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 230 Oct 15 2025 Ewhile.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 230 Oct 15 2025 Ewhile.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 991 Nov 12 2025 FactorialDemo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 287 Dec 2 2025 FactorialDemo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1194 Oct 13 2025 Floating.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 754 Oct 14 2025 Grades.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 409 Oct 14 2025 Grades.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 362 Oct 14 2025 'Int variable.java'
-rwxrwxrwx 1 keerthi_priya keerthi_priya 362 Oct 14 2025 'Int variables.java'
-rwxrwxrwx 1 keerthi_priya keerthi_priya 597 Dec 2 2025 Jhansi.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 251 Dec 2 2025 Jhansi.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 95 Jan 5 2026 Keerthi.html

```

```

keerthi_priya@LAPTOP-OSGV  ×  +  ▾
-rwxrwxrwx 1 keerthi_priya keerthi_priya 696 Oct 13 2025 Postive.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 555 Oct 15 2025 Print.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1079 Oct 13 2025 ScanDemo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 263 Oct 13 2025 ScanDemo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 619 Oct 22 2025 StringDemo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 418 Oct 22 2025 StringDemo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1039 Oct 16 2025 SubMul.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 344 Oct 16 2025 SubMul.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 953 Oct 22 2025 SumDemo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 225 Oct 15 2025 SumDemo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 244 Oct 22 2025 SumDemo3.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1658 Nov 11 2025 SumMatrix.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 865 Nov 11 2025 SumMatrix.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 717 Oct 23 2025 Sumeven.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 299 Oct 23 2025 Sumeven.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1157 Oct 23 2025 Sumodd.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 318 Oct 23 2025 Sumodd.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 335 Nov 26 2025 TestDemo.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 282 Nov 26 2025 TestDemo.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 845 Oct 14 2025 Three.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 474 Oct 14 2025 Three.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 372 Nov 26 2025 child.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 301 Oct 13 2025 even.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 373 Oct 13 2025 floating.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 721 Sep 17 12:10 p-6.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 4096 Dec 25 2025 p1
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16248 Sep 17 12:15 p62
-rwxrwxrwx 1 keerthi_priya keerthi_priya 386 Nov 26 2025 parent.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 264 Oct 15 2025 print.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16448 Sep 17 15:21 session3_task2
-rwxrwxrwx 1 keerthi_priya keerthi_priya 3108 Sep 17 15:06 session3_task2.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16184 Sep 17 15:34 session4_task1
-rwxrwxrwx 1 keerthi_priya keerthi_priya 2249 Sep 17 15:44 session4_task1.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16272 Sep 17 15:45 session4_task2
-rwxrwxrwx 1 keerthi_priya keerthi_priya 2238 Sep 17 15:52 session4_task2.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16184 Sep 17 15:54 session5_task1
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1551 Sep 17 15:54 session5_task1.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16312 Sep 17 16:23 session5_task2
-rwxrwxrwx 1 keerthi_priya keerthi_priya 3340 Sep 17 16:23 session5_task2.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16184 Sep 17 16:31 session6_task1
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1205 Sep 17 16:31 session6_task1.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 16352 Sep 17 16:38 session6_task2
-rwxrwxrwx 1 keerthi_priya keerthi_priya 1200 Sep 17 16:38 session6_task2.c
-rwxrwxrwx 1 keerthi_priya keerthi_priya 312 Dec 18 2025 vinnu.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 172 Dec 18 2025 vinnu.java
-rwxrwxrwx 1 keerthi_priya keerthi_priya 757 Oct 16 2025 vote.class
-rwxrwxrwx 1 keerthi_priya keerthi_priya 370 Oct 16 2025 vote.java

```

```

Child exited with status: 0
Parent process completed.

```